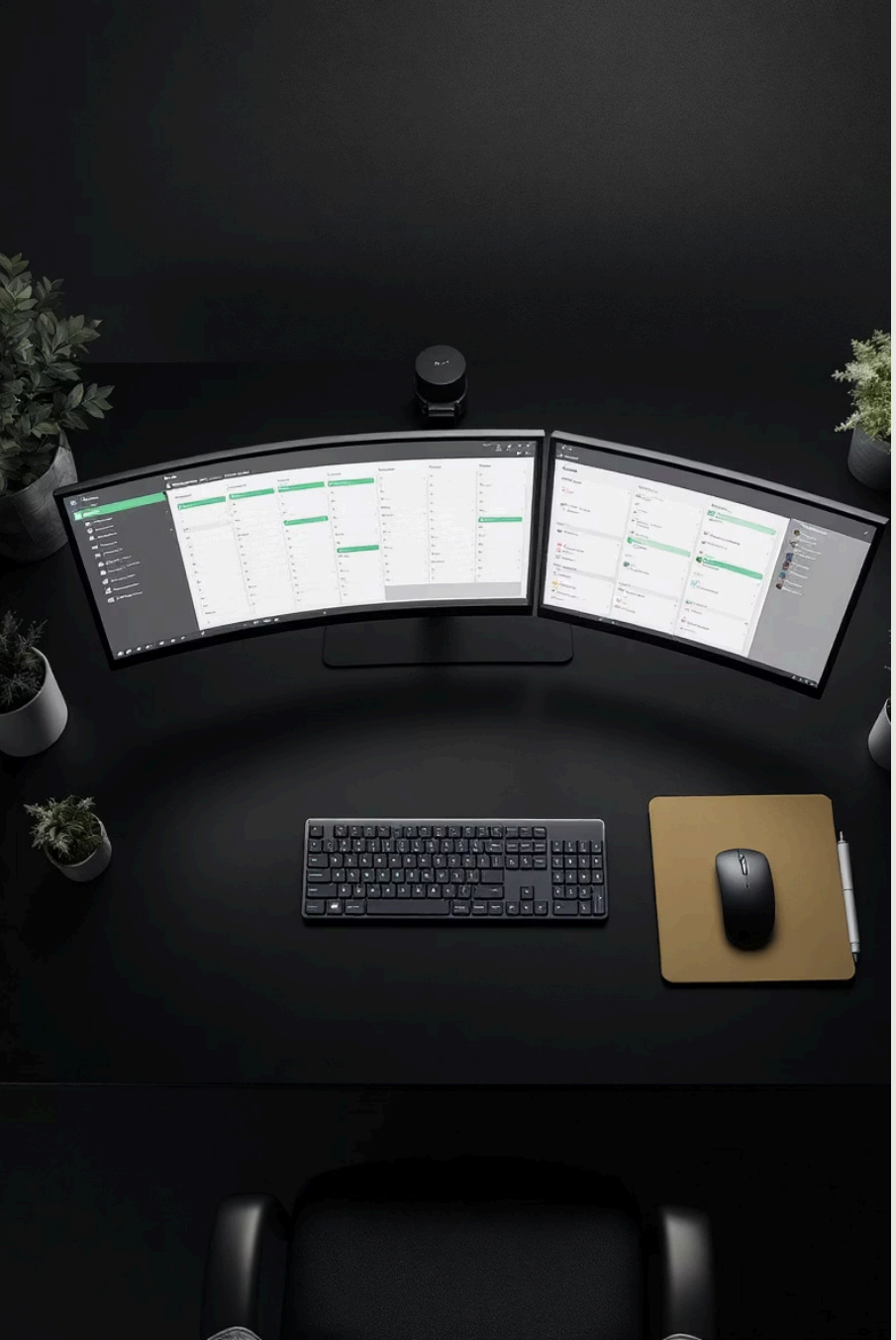

DAILY PRO — SMART HABIT TRACKING SYSTEM

A lightweight, scalable habit-management app built with React and TypeScript. Calendar-first design, local persistence, and developer-friendly architecture for rapid iteration.

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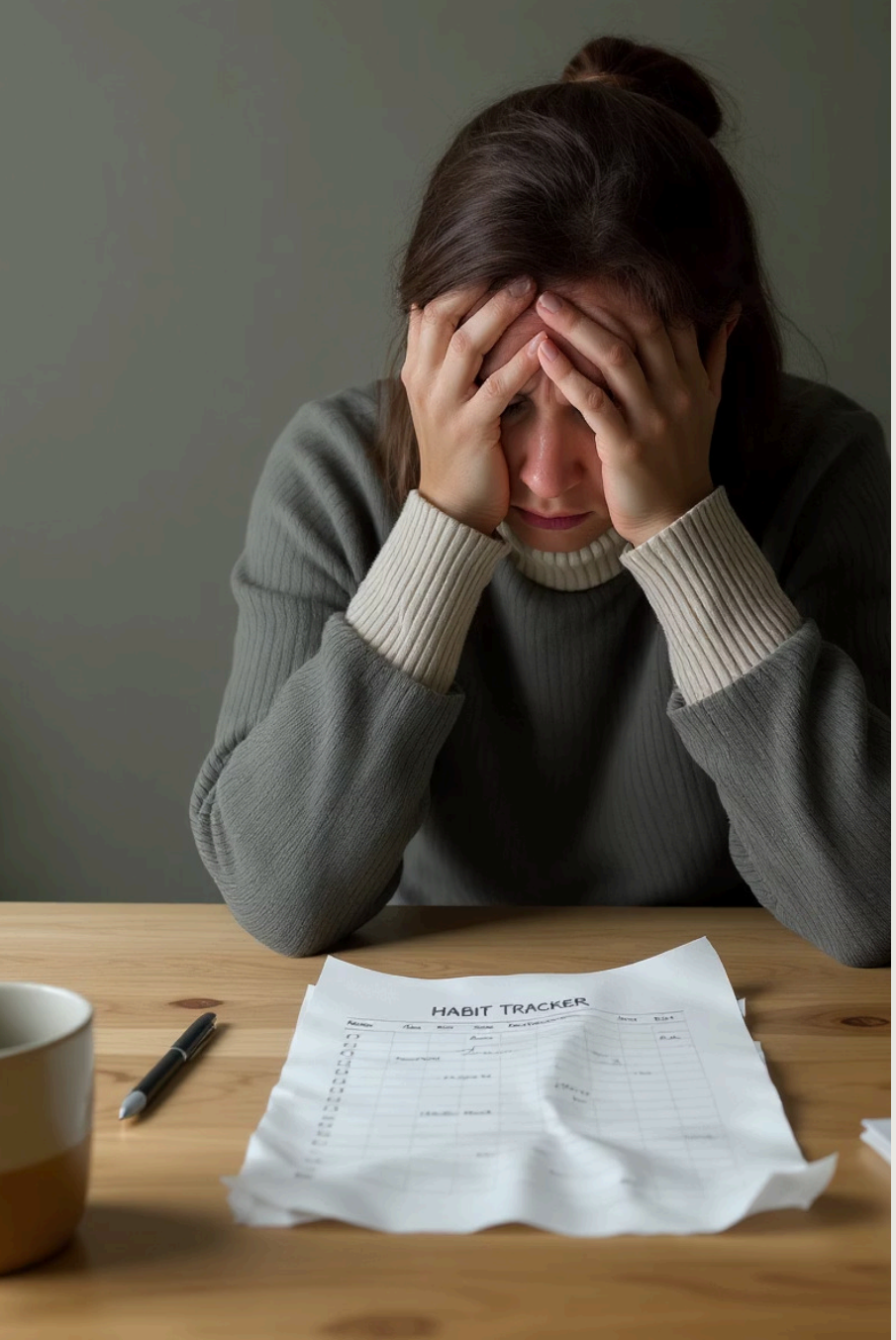

PROBLEM STATEMENT

DAILY FRICTION

People struggle to maintain habits without clear, immediate feedback.

POOR MOTIVATION

Limited visualization reduces engagement and momentum.

INCONSISTENT TRACKING

Manual systems are error-prone and scattered across tools.

NO CENTRALISED VIEW

Habits, mood and progress rarely live together in a single UI.

Solution: Daily Pro delivers a structured, visual and interactive calendar that combines habit completion and mood tracking in one place.

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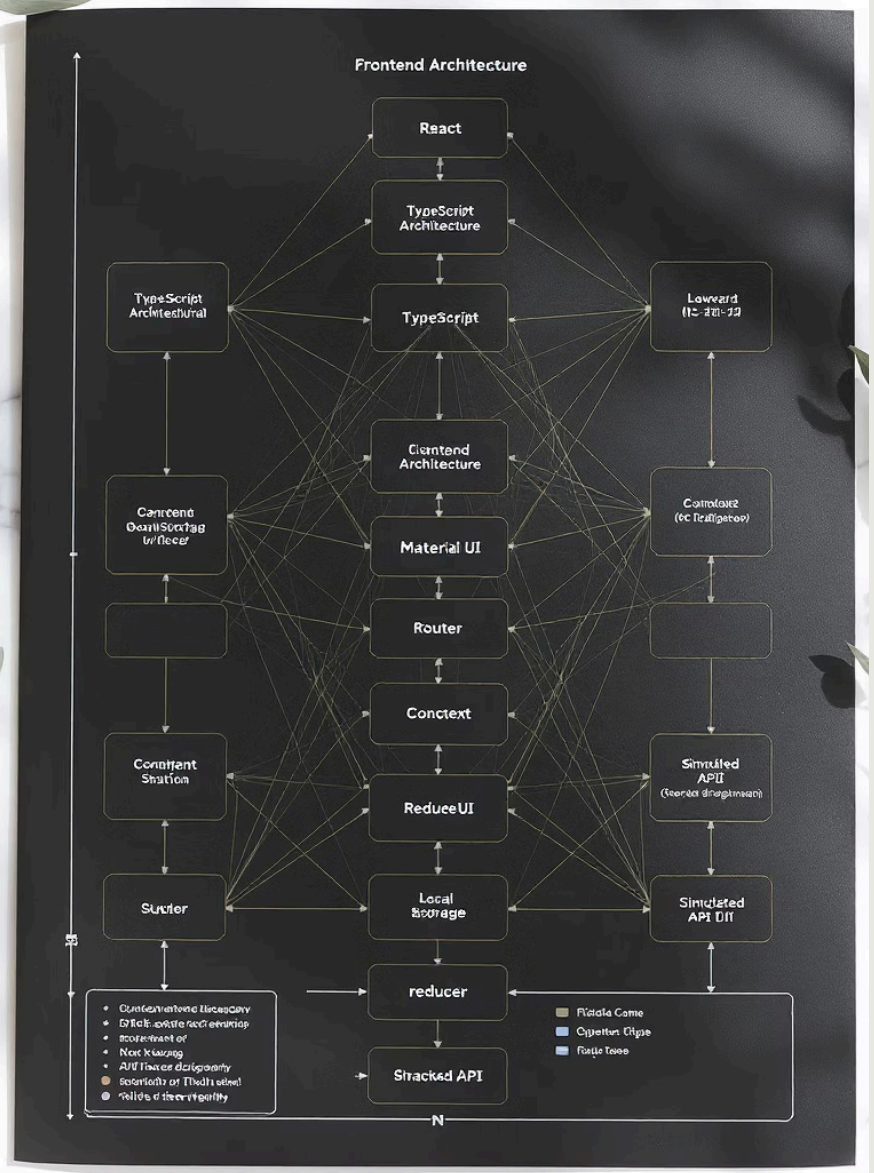

PROJECT ARCHITECTURE – FRONTEND FOCUS

FRONTEND STACK

- React (functional components)
- TypeScript for strict types and DX
- Material UI for consistent components
- React Router for navigation

STATE & DATA

- useReducer for centralised state machine
- React Context to expose auth & calendar state
- LocalStorage for month-wise persistence
- Simulated API fetch during initial load

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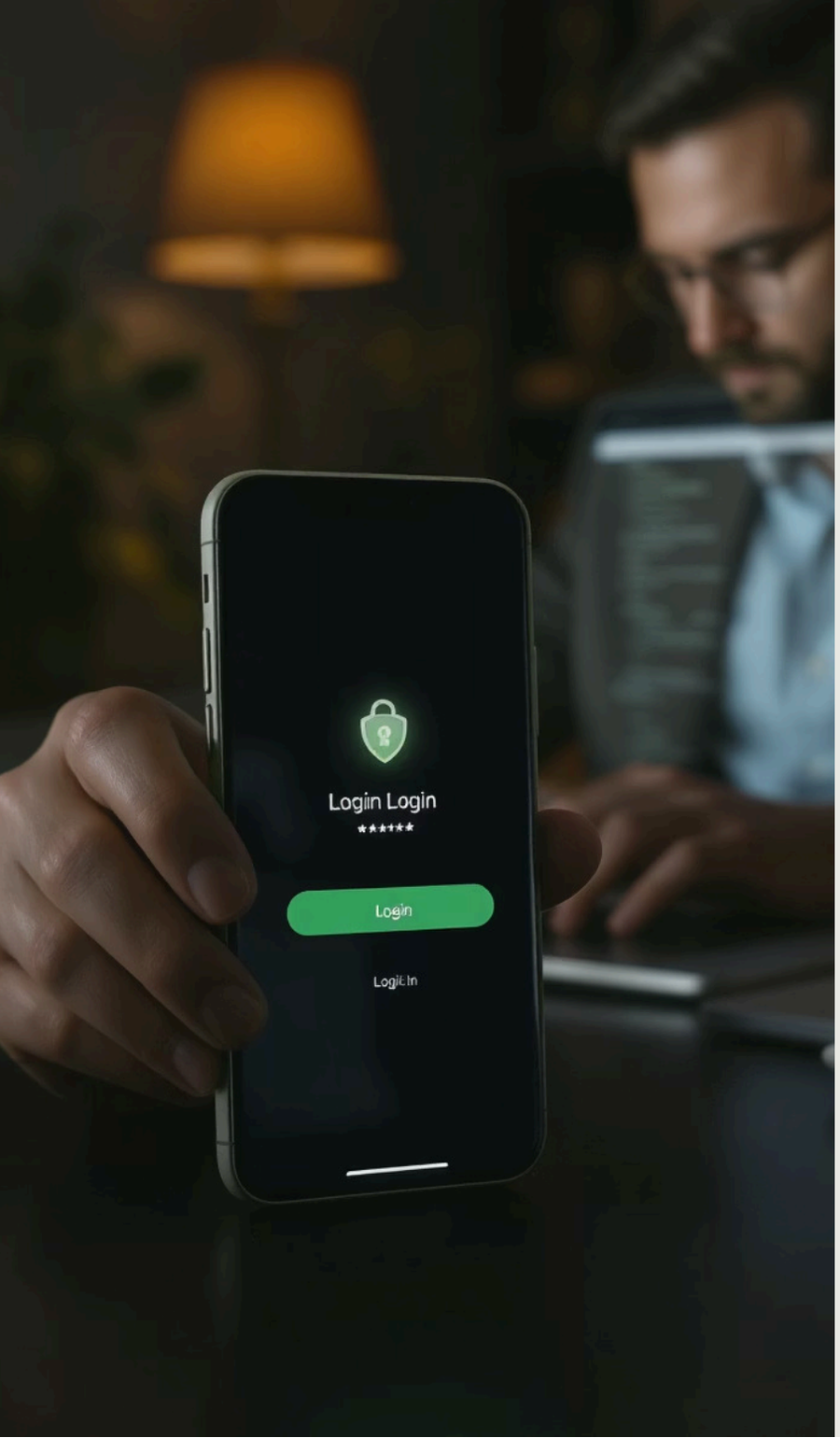

AUTHENTICATION & PROTECTED ROUTES

AUTH MODEL

AuthContext stores user state and exposes login/logout actions.

PERSISTENCE

Auth state is mirrored to LocalStorage to survive refreshes.

ROUTE PROTECTION

ProtectedRoute (HOC) redirects unauthenticated users to login.

DEVELOPER API

useAuth custom hook simplifies access to auth actions and state.

Key files: AuthContext.tsx, ProtectedRoute.tsx, useAuth.ts — minimal, typed and easy to test.

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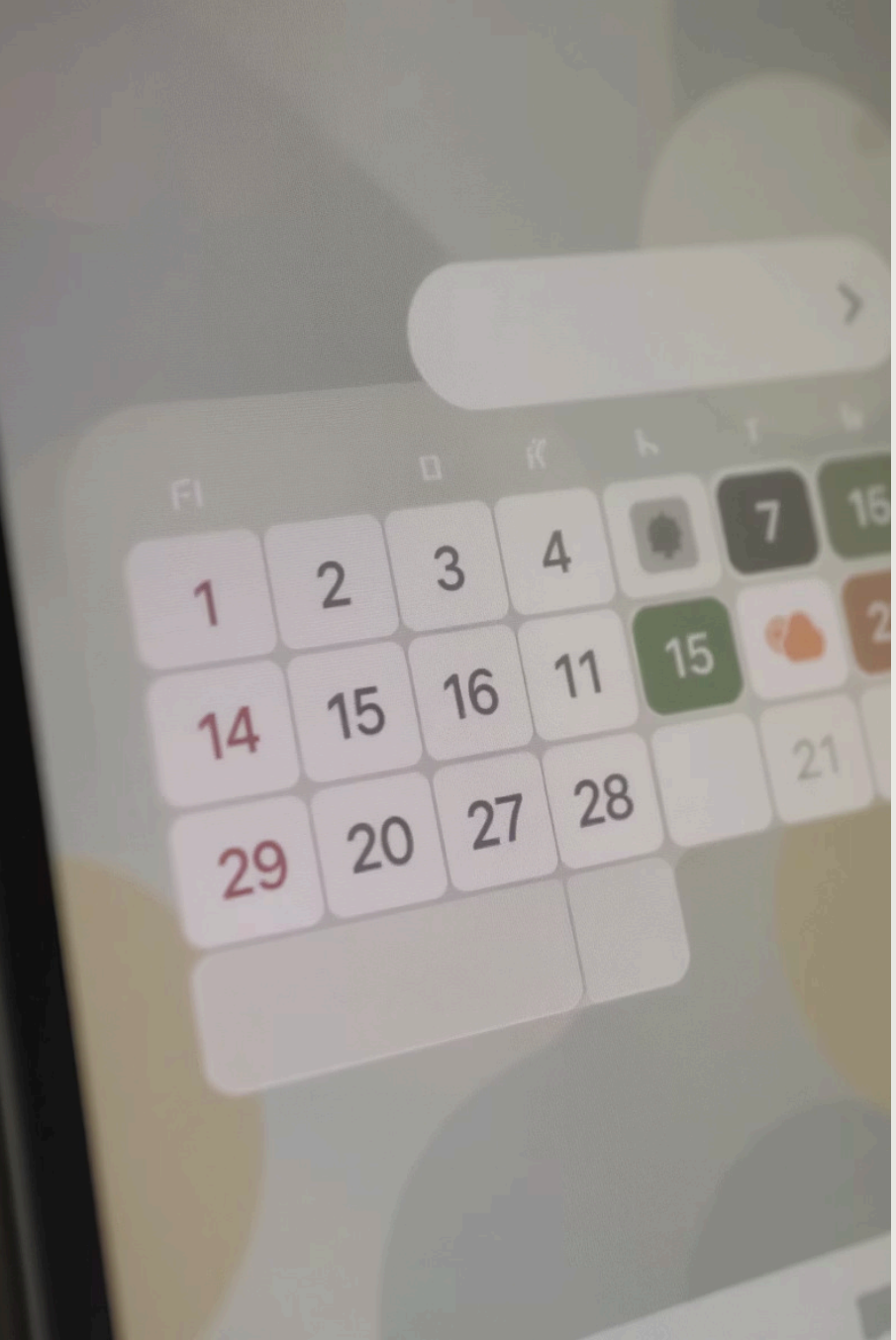

CALENDAR STATE MANAGEMENT — USEREDUCER

The calendar uses a single reducer to manage complex interactions across months, habits and moods. Centralised actions make transitions explicit and easier to test.

01

STRUCTURED ACTIONS

SET_DATA, ADD_HABIT, TOGGLE_HABIT, SET_MOOD — clear intent per action.

02

PREDICTABLE STATE

Pure reducer logic enables time-travel testing and snapshot stability.

03

SCALABLE

New actions (analytics, import/export) integrate cleanly into the reducer.

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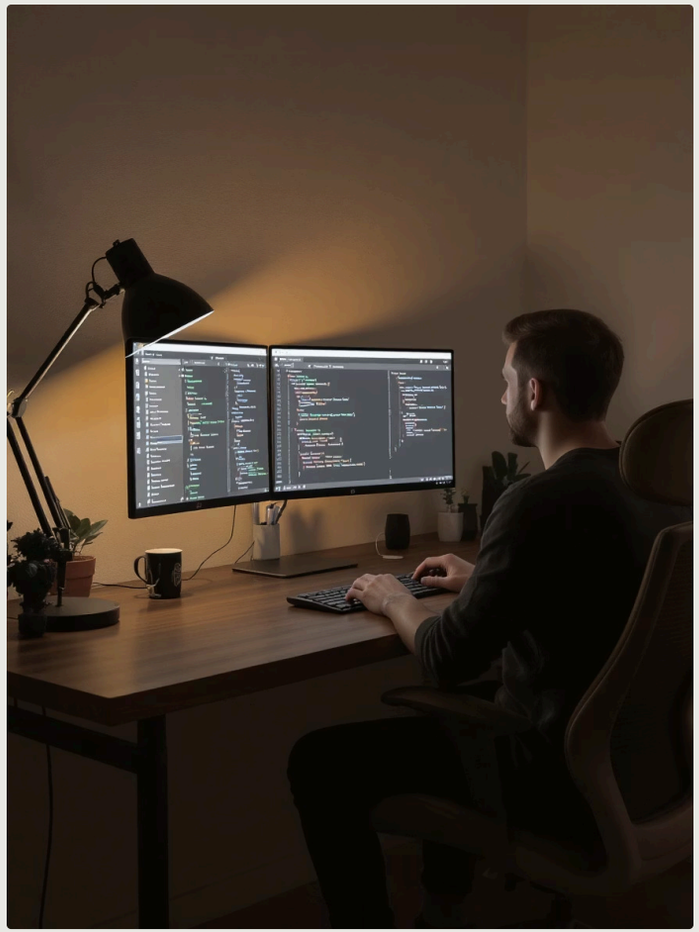
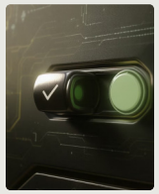


HABIT TRACKING LOGIC — COMPONENTS

Core UI is modular and testable. Components are small, typed and single-responsibility.

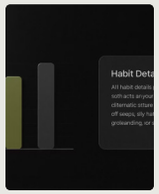


HABITCELL.TSX

Renders day-level state and toggles completion with optimistic UI.

MOODSELECTOR.TSX

Small, accessible control for daily mood selection persisted to the reducer.



HABITDETAILS.TSX

Detail panel for habit history, edit and remove actions.

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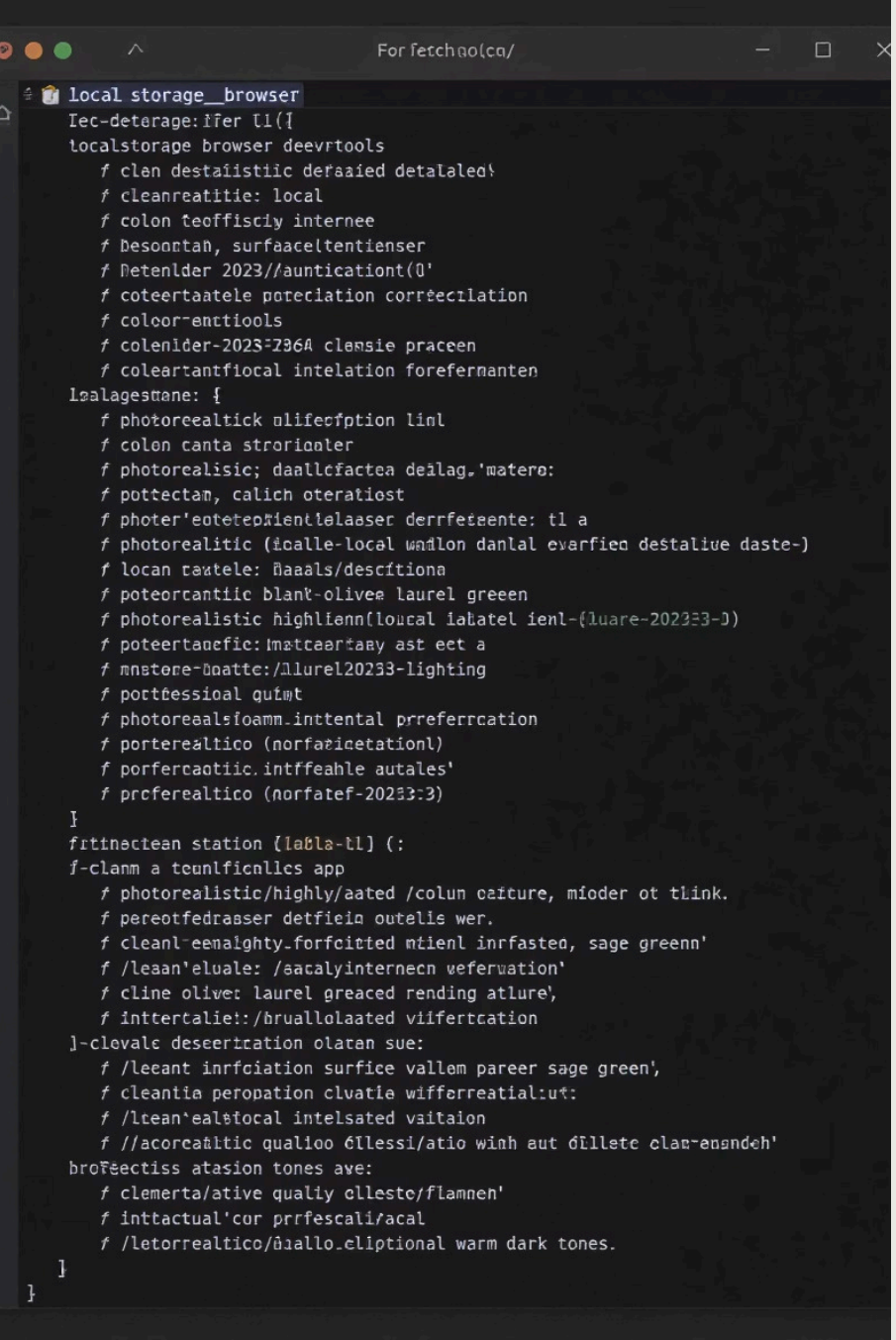

DATA FETCHING & PERSISTENCE

The app loads data asynchronously on mount (simulated fetch) and writes month-scoped state to LocalStorage using keys of the form: `calendar-year-month`.

BENEFITS

- Fast client-only performance
- Offline-friendly behaviour
- No backend required for MVP

CONSIDERATIONS

- Key rotation for migrations
- Sync conflicts if multi-device required (future work)

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PERFORMANCE OPTIMISATION

- ## MEMOISATION

  useMemo to memoise context value and derived calendar data.

- ## REDUCER STRUCTURE

  Batch updates and minimal payloads to reduce diff work.

- ## RENDER CONTROL

  Pure components and key-based lists to avoid unnecessary re-renders.

- ## EFFICIENT DATE GENERATION

  Cached month matrices and lazy generation for neighbouring months.

Result: smooth month navigation, snappy toggles, and predictable UX even with many habits.

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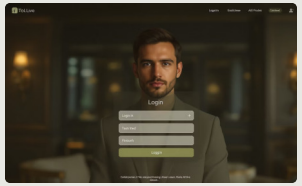
KEY FEATURES — AT A GLANCE

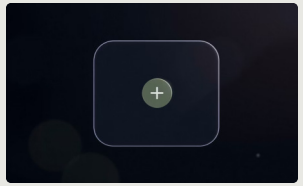


LOGIN & LOGOUT

Simple auth flow with LocalStorage-backed session.

MONTHLY CALENDAR VIEW

Visual, month-scoped habit overview with per-day toggles.

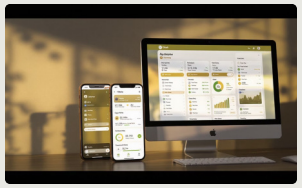


DYNAMIC HABIT ADDITION

Add habits at runtime — they populate existing month days automatically.

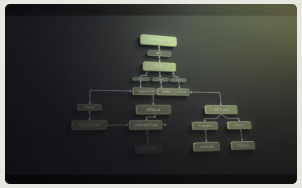
MOOD TRACKING

Associate mood with each day to capture qualitative context.



LOCALSTORAGE PERSISTENCE

Fast, offline-capable persistence without server dependency.



CLEAN COMPONENT ARCHITECTURE

Small, single-responsibility components with strong typing.

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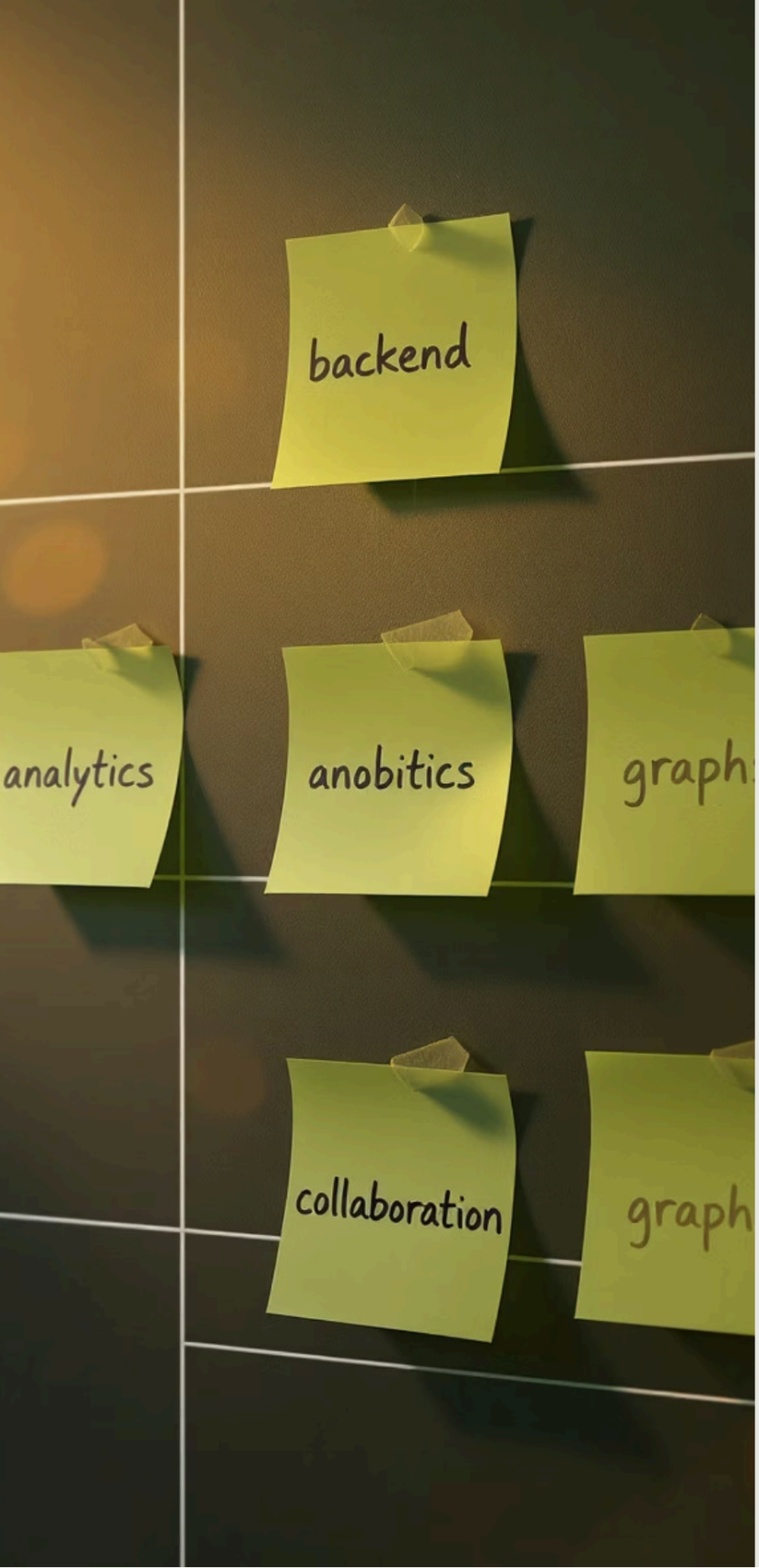

CONCLUSION & FUTURE ENHANCEMENTS

Daily Pro is a scalable, client-first habit tracker implemented with professional React patterns. It provides a solid foundation for growth.

BACKEND API INTEGRATION

Enable cross-device sync and secure user storage.

PER-USER STORAGE

User-scoped accounts and server-side persistence.

ANALYTICS DASHBOARD

Graph-based insights: streaks, completion rates, mood correlations.

MOBILE APP

Native mobile experience and push reminders to increase retention.

Next steps for engineers: extract reducer tests, add integration tests for ProtectedRoute, and prototype a small analytics microservice to ingest monthly snapshots.

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